EXPERIMENT 3

Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.

1. AIM

To design a flowchart prototype in Figma that illustrates the operation of the Network Switching Subsystem (NSS) in a mobile cellular system, including call initiation, subscriber authentication, call routing, connection establishment, and call termination.

2. SOFTWARE USED

- Software: Figma
- Platform: Web-based design tool
- Tools: Shapes, text, connectors, and arrows
- Output: Network Switching Subsystem flowchart prototype

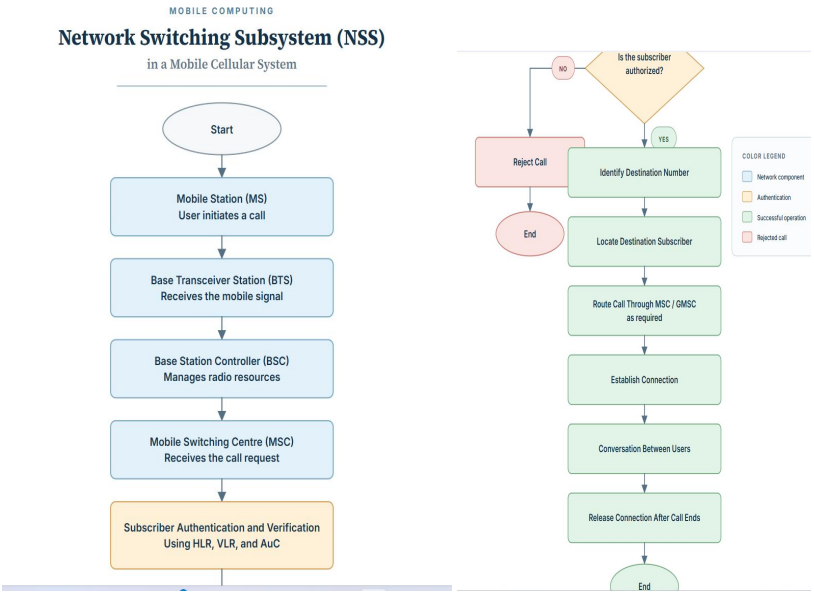
3. ALGORITHM

1. Start the process.
2. The Mobile Station (MS) initiates a call.
3. The BTS receives the signal, and the BSC manages radio resources.
4. The MSC receives the call request.
5. Verify the subscriber using HLR, VLR, and AuC.
6. If the subscriber is not authorized, reject the call and terminate the process.
7. If authorized, identify the destination subscriber and route the call.
8. Establish the connection between the users.
9. Allow communication until the call ends.
10. Release the connection and terminate the process.

4. DESIGN

The flowchart is designed using standard flowchart symbols:

Symbol	Purpose
Oval	Start and End
Rectangle	Processing steps
Diamond	Authentication decision
Arrow	Direction of process flow



5. PROTOTYPE

The prototype represents the call-processing operation of the Network Switching Subsystem in a mobile cellular network.

- Input: Call request from a mobile station.
- Processing: Subscriber authentication, destination identification, and call routing.
- Output: Successful call connection or call rejection.

The prototype uses a light-blue and green color scheme, with orange for authentication and red for call rejection. Arrows connect the steps, and the decision diamond contains clearly labeled YES and NO branches.

Result: A flowchart prototype is designed in Figma to demonstrate the working of the Network Switching Subsystem in a mobile cellular system.